Math 420 - Wed. 11/15

Diagonalization: An $n \times n$ matrix A is diagonalizable if $\exists$ an invertible $n \times n$ matrix Q and diagonal matrix D s.t.

$$A = \underbrace{Q}_{\hookrightarrow\ n \text{ linearly independent eigenvectors}} D Q^{-1}$$

If $\lambda_1 \neq \lambda_2 \neq \dots \neq \lambda_n$, then there will be n linearly independent eigenvectors.

What if $\lambda_j = \lambda_i$ $\quad 1 \leq i \neq j \leq n$?

Algebraic multiplicity = the # of times an eigenvalue gets repeated

Geometric multiplicity = the # of corresponding linearly indep. eigenvectors.

If a.m. = g.m then A is diagonalizable

If g.m. < a.m. A is not diagonalizable $\rightarrow$ Jordan Normal Form (§6.1)

If we add that the matrix A is symmetric or Hermitian.

$\hookrightarrow A = A^*$ $\qquad$ $\hookrightarrow A = A^T$ (real)

<u>Claim</u> Symmetric/Hermitian matrices have all real eigenvalues.

<u>Proof</u> Suppose A is Hermitian. Then $A = A^*$.

~~Let λ be an eigenvalue of A with x,~~

$Ax = \bar{\lambda} x$ $\qquad$ $Ax = \lambda x \rightarrow x^* A x = x^* \lambda x$

$\Rightarrow \lambda = \bar{\lambda}$ $\qquad$ $(Ax)^* x = x^* A^* x$ $\qquad$ $\lambda x^* x = (\lambda x^*)^*$

Hermitian $A = A^*$ $\qquad \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} x = \lambda x$

<u>Claim</u> The eigenvectors corresponding to different eigenvalues of a Hermitian (symmetric) matrix are orthogonal!

<u>Proof</u> A is Hermitian with $\lambda \neq \mu$ eigenvalues and $Ax = \lambda x$, $Ay = \mu y$.

WTS that $x \perp y \Rightarrow \langle x, y \rangle = 0 = \langle y, x \rangle$

$$\begin{array}{c} x^* \\ \underset{1 \times n}{[\quad]} \end{array} \begin{array}{c} y \\ \underset{n \times 1}{\begin{bmatrix} \\ \\ \\ \end{bmatrix}} \end{array} = 1 \times 1$$

$$\begin{aligned} \| \; & \cancel{x^* y = 0 = y^* x} \; \| \\ & y^* x = 0 = x^* y \end{aligned}$$

$$y^* (Ax = \lambda x) \Rightarrow y^* A x = y^* \lambda x$$

$$\begin{aligned} y^* A x &= \lambda y^* x \quad \Big\} \; A = A^* \\ y^* A^* x &= \lambda y^* x \\ (Ay)^* x &= \lambda y^* x \\ (\mu y)^* x &= \lambda y^* x \\ \mu y^* x = \bar{\mu} y^* x &= \lambda y^* x \end{aligned}$$

$$\begin{aligned} \mu y^* x - \lambda y^* x &= 0 \\ \underbrace{(\mu - \lambda)}_{\neq 0} (y^* x) &= 0 \\ \hookrightarrow \quad y^* x &= 0 \quad x \perp y \quad \square \end{aligned}$$

$A = QDQ^{-1}$ $\rightsquigarrow$ but we know all col's of Q are orthogonal

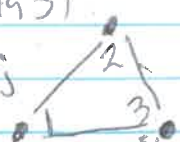
$Q^{-1} = Q^T$

A quantum walk is defined by a transition matrix on a graph (M453)

$\|$

$U(t) = e^{itA}$.

$$A = [a]_{ij} = \begin{cases} 1 & i \leftrightarrow j \\ 0 & i \not\leftrightarrow j \end{cases}$$



$$\begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

K_3 = complete graph on 3 vertices

$$A = \begin{array}{c|ccc} & 1 & 2 & 3 \\ \hline 1 & 0 & 1 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 1 & 1 & 0 \end{array}$$

The adjacency matrix is symmetric!
$\Rightarrow$ eigenvalues are real
eigenvectors orthogonal

$$U(t) = e^{itA}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$U(t) = \sum_{n=0}^{\infty} \frac{(itA)^n}{n!}$$

U unitary & symmetric.

E_{λ_i} = orthogonal proj onto λ_i eigenspace

$$A = \sum_j \lambda_j E_{\lambda_j}$$

$$U(t) = \sum_j e^{it\lambda_j} E_{\lambda_j}$$